

A Literature-Implied Counterexample to L^∞ -BIBO/Admissibility Equivalence

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Status: full negative literature-implied answer. Kato's diagonal semigroup on c_0 , recorded in Example 2.3 of arXiv:2008.00459, makes the identity-output system L^∞ -BIBO stable although its control operator is not L^∞ -admissible. This answers Remark 5.9 of arXiv:2408.06313 negatively. The construction itself is known literature; the relation to the later question is the identification recorded here.

1 Question and answer

Let A generate a C_0 -semigroup on a Banach space X , let $B \in \mathcal{L}(U, X_{-1})$, and consider the identity-output system node

$$\Sigma_B = \Sigma(A, B, I, (\cdot I - A)^{-1}B).$$

Schwenninger and Wierzbza prove in Proposition 5.8 of [1] that Σ_B is C^∞ -BIBO stable if and only if B is infinite-time C -control-admissible. On arXiv PDF page 12 they ask:

Source question (Remark 5.9). *Does the analogous equivalence hold for L^∞ inputs?*

Literature-implied answer. *No. There is an exponentially stable positive diagonal semigroup on c_0 and a positive control operator B for which Σ_B is L^∞ -BIBO stable with constant one, while B is not even finite-time L^∞ -control-admissible.*

2 The Kato system

Let $X = U = c_0(\mathbb{N})$ and define

$$(T(t)x)_n = e^{-nt}x_n, \quad (Ax)_n = -nx_n,$$

with

$$D(A) = \{x \in c_0 : (nx_n)_{n \geq 1} \in c_0\}.$$

Then $(T(t))_{t \geq 0}$ is a positive exponentially stable C_0 -semigroup. Using $0 \in \rho(A)$ to norm the extrapolation space,

$$X_{-1} = \{z = (z_n) : (z_n/n) \in c_0\}, \quad \|z\|_{-1} = \sup_n \frac{|z_n|}{n}.$$

Set $B = -A_{-1}|_X$, so $(Bu)_n = nu_n$. This is a bounded positive map $c_0 \rightarrow X_{-1}$. Also

$$((\lambda I - A)^{-1}Bu)_n = \frac{n}{\lambda + n}u_n \in c_0,$$

so Σ_B is a legitimate system node with output space c_0 . Its combined observation/feedthrough operator is the identity: for every admissible pair (x, u) ,

$$(C\&D)\begin{pmatrix} x \\ u \end{pmatrix} = x - (\lambda I - A)^{-1}Bu + (\lambda I - A)^{-1}Bu = x.$$

Consequently its distributional output is the state itself.

For $u \in L_{\text{loc}}^\infty([0, \infty); c_0)$ and zero initial state, the extrapolation-space state has coordinates

$$x_n(t) = n \int_0^t e^{-n(t-s)} u_n(s) ds. \quad (1)$$

3 L^∞ -BIBO stability

The following elementary verification makes the system-node identification with the maximal-regularity statement in [2] explicit.

Proposition 1. *For every $u \in L_{\text{loc}}^\infty([0, \infty); c_0)$, the state in (1) has a c_0 -valued strongly measurable representative and, for every $\tau > 0$,*

$$\|x\|_{L^\infty([0, \tau]; c_0)} \leq \|u\|_{L^\infty([0, \tau]; c_0)}.$$

Thus Σ_B is L^∞ -BIBO stable.

Proof. The coordinate estimate from (1) gives, for almost every $t \leq \tau$,

$$|x_n(t)| \leq (1 - e^{-nt})\|u\|_{L^\infty([0, t]; c_0)} \leq \|u\|_{L^\infty([0, \tau]; c_0)}.$$

It remains to prove that $x_n(t) \rightarrow 0$ for almost every t . Extend u by zero to negative times and take a Lebesgue point t of the Bochner function u . Writing $r = t - s$ gives

$$x_n(t) = (1 - e^{-nt})u_n(t) + n \int_0^t e^{-nr} (u_n(t - r) - u_n(t)) dr.$$

The first term tends to zero because $u(t) \in c_0$. The absolute value of the second is bounded by

$$n \int_0^t e^{-nr} \|u(t - r) - u(t)\|_{c_0} dr,$$

which tends to zero because $(ne^{-nr})_{n \geq 1}$ is a one-sided approximate identity and t is a Lebesgue point. Hence $x(t) \in c_0$ for almost every t .

Each coordinate is measurable. On the full-measure set just obtained, finite-coordinate truncations converge pointwise in c_0 norm to $x(t)$; therefore x has a strongly measurable c_0 -valued representative. The coordinate estimate then proves the asserted essential-supremum bound. $\square$

The supporting paper states the same regularity fact in its original language: immediately after Example 2.3, it records that Kato's semigroup has L^∞ -maximal regularity. Since $B = -A_{-1}$, that assertion is exactly the a.e. c_0 -valued boundedness of the state above.

4 Failure of endpoint admissibility

Fix $\tau > 0$. Example 2.3 of [2], on arXiv PDF pages 7–8, uses the strongly measurable input $u \in L^\infty([0, \tau]; c_0)$ given by

$$u_n(s) = \begin{cases} 1, & s \in [\tau - 1/n, \tau - 1/(2n)] \text{ and } 1/n < \tau, \\ 0, & \text{otherwise.} \end{cases}$$

For each fixed s , only finitely many coordinates are nonzero, so $u(s) \in c_0$, and $\|u\|_{L^\infty} = 1$. But at the endpoint,

$$x_n(\tau) = \int_{1/(2n)}^{1/n} n e^{-nr} dr = e^{-1/2} - e^{-1} > \frac{1}{5}$$

for every sufficiently large n . Thus $x(\tau) \notin c_0$. The endpoint input map

$$\Phi_\tau u = \int_0^\tau T_{-1}(\tau - s) B u(s) ds$$

does not have range in X , so B is not L^∞ -control-admissible. Together with Proposition 1, this is the desired counterexample.

The mechanism is sharp: BIBO stability only sees the output as an almost- everywhere equivalence class, whereas control admissibility requires the state to lie in X at the specified endpoint. Kato's input creates a single non- c_0 endpoint without violating the a.e. BIBO estimate.

5 Reflexive-state boundary

The endpoint defect cannot occur on reflexive state spaces.

Proposition 2. *If X is reflexive, then for the identity-output system Σ_B , L^∞ -BIBO stability is equivalent to infinite-time L^∞ -control-admissibility.*

Proof. Only the forward implication needs comment. Fix $\tau > 0$ and $u \in L^\infty([0, \tau]; U)$, extended by zero after τ . Let x be the zero-state trajectory. BIBO stability gives $x(t) \in X$ for almost every $t < \tau$ and $\|x(t)\|_X \leq c\|u\|_\infty$. Choose $t_k \uparrow \tau$ from this full-measure set. Reflexivity gives a subsequence converging weakly in X to some $z \in X$. The canonical embedding $X \hookrightarrow X_{-1}$ is continuous, while the generalized state is continuous in X_{-1} ; hence the same subsequence converges in X_{-1} norm to $x(\tau)$. Uniqueness of weak limits in X_{-1} gives $x(\tau) = z \in X$, with $\|x(\tau)\|_X \leq c\|u\|_\infty$. The constant is independent of τ , which is infinite-time admissibility. The converse follows directly from the state formula and the identity output. $\square$

This addendum is not needed for the negative answer, but it isolates the role of the nonreflexive space c_0 .

6 Provenance and remaining scope

The semigroup, the operator $B = -A_{-1}$, the explicit bad input, the L^∞ -maximal-regularity property, and the failure of L^∞ -admissibility are all contained in Example 2.3 and the following paragraph of [2]. The present packet makes the direct translation to Remark 5.9 of [1]; no new counterexample is claimed.

This does not settle the broader Outlook question in [1] asking whether C^∞ -, C -, and L^∞ -BIBO stability are equivalent for arbitrary system nodes. In the Kato identity-output example all three BIBO properties hold; what fails is endpoint L^∞ control admissibility.

References

- [1] F. L. Schwenninger and A. A. Wierzba, *A dual notion to BIBO stability*, J. Math. Anal. Appl. **545** (2025), 129156; arXiv:2408.06313.
- [2] B. Jacob, F. L. Schwenninger, and J. Wintermayr, *A refinement of Baillon's theorem on maximal regularity*, Stud. Math. **263** (2022), 141–158; arXiv:2008.00459.